

## Exercise 2

1. A patient goes to see a doctor. Suppose the doctor performs a test with 99% reliability, i.e., 99% of people who are sick test positive and 99% of the healthy people test negative. The doctor knows that only 1% of the people in the country are sick. If a patient has tested positive, what is the probability he is sick?
2. How does a Naïve Bayes classifier work? Explain using the underlying formula. Why is it “naïve”?
3. How does the Naïve Bayes classifier handle numeric attributes? Give the formula.
4. For the dataset provided in Table 1, compute the conditional probabilities of all attribute values, given the target attribute Defaulted Borrower (DB=Y and DB=N). Classify the following instance according to the the Naïve Bayes Classifier without handling zero conditional probabilities:

$$X = (\text{Home Owner} = N, \text{Marital Status} = \text{Married}, \text{Annual Income} = 90)$$

5. Explain the zero-frequency problem. How can it be avoided?

For the dataset provided in Table 1, calculate the conditional probabilities of all attribute, given the target attribute Defaulted Borrower (DB=Y and DB=N) using Laplace estimate and then classify the following instance:

$$X = (\text{Home Owner} = N, \text{Marital Status} = \text{Married}, \text{Annual Income} = 90)$$

Table 1: Training set for defaulted borrower classification.

| Home<br>Owner | Marital<br>Status | Annual<br>Income | Defaulted<br>Borrower |
|---------------|-------------------|------------------|-----------------------|
| Y             | Single            | 125              | N                     |
| N             | Married           | 100              | N                     |
| N             | Single            | 70               | N                     |
| Y             | Married           | 120              | N                     |
| N             | Divorced          | 95               | Y                     |
| N             | Married           | 60               | N                     |
| Y             | Divorced          | 220              | N                     |
| N             | Single            | 85               | Y                     |
| N             | Married           | 75               | N                     |
| N             | Single            | 90               | Y                     |